Hints: Combinatorics – 2. Recurrences and generating functions

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Multiply $A(x)$ by $Q(x) = 1 - c_1x - \cdots - c_kx^k$: the coefficient of x^n in $Q(x)A(x)$ is $a_n - c_1a_{n-1} - \cdots - c_ka_{n-k}$, which vanishes for $n \geqslant k$. So QA is a polynomial of degree $< k$, and conversely.

Hint 2. For the explicit form, decompose P/Q into partial fractions $\sum_i \sum_{j<m_i} \frac{b_{ij}}{(1-\lambda_i x)^{j+1}}$ and use $[x^n](1-\lambda x)^{-(j+1)} = \binom{n+j}{j}\lambda^n$, a polynomial in n of degree j times λ^n. For $a_n = 4a_{n-1} - 4a_{n-2}$ the root 2 is double, so $a_n = (\alpha + \beta n)2^n$; the initial values give $a_n = (n+2)2^{n-1}$.

Hint 3. If all $|\lambda_i| < 1$, then $a_n = \sum p_i(n)\lambda_i^n \to 0$; integers tending to 0 are eventually 0. (Once k consecutive terms vanish, the recurrence keeps all later ones at 0 anyway.)

Problem 2

Hint 1. Solve $(\sum a_nx^n)(\sum b_nx^n) = 1$ coefficient by coefficient: $a_0b_0 = 1$ and $a_0b_n = -\sum_{k\geqslant 1} a_kb_{n-k}$ determine the b_n recursively exactly when $a_0 \neq 0$.

Hint 2. For $1/(1-x-x^2) = \sum b_nx^n$ the recursion gives $b_n = b_{n-1} + b_{n-2}$, $b_0 = b_1 = 1$, so $b_n = F_{n+1}$. The dictionary rules: a pair of objects of sizes i, j contributes x^{i+j}, so the generating function of pairs is the product; a sequence of m pieces is counted by F^m, and $\sum_m F^m = 1/(1-F)$ is a well-defined formal series when $F(0) = 0$.

Hint 3. Compositions into parts 1 and 2: sequences of pieces counted by $F = x + x^2$, so the generating function is $1/(1-x-x^2)$ and the count is F_{n+1}.

Problem 3

Hint 1. Compute a few more terms or test a linear recurrence of order 2 on the given ones: $26 = 6 \cdot 5 - 4 \cdot 1$, $136 = 6 \cdot 26 - 4 \cdot 5$, … The roots of $t^2 - 6t + 4$ are $3 \pm \sqrt{5}$.

Hint 2. Define $y_n = A(3+\sqrt{5})^n + B(3-\sqrt{5})^n$ with $y_0 = 1$, $y_1 = 5$, i.e. $A = \frac{5+2\sqrt{5}}{10}$, $B = \frac{5-2\sqrt{5}}{10}$. Check that $y_{n+1} - 3y_n = \sqrt{5}\,(A\alpha^n - B\beta^n)$ while $\sqrt{5}\,y_n = \sqrt{5}\,(A\alpha^n + B\beta^n)$.

Hint 3. So $\sqrt{5}\,y_n - (y_{n+1} - 3y_n) = 2\sqrt{5}\,B(3-\sqrt{5})^n \in (0,1)$, hence $y_{n+1} = 3y_n + \lfloor y_n\sqrt{5} \rfloor$ and $x_n = y_n = \frac{5+2\sqrt{5}}{10}(3+\sqrt{5})^n + \frac{5-2\sqrt{5}}{10}(3-\sqrt{5})^n$ by induction.

Problem 4

Hint 1. Look for a polynomial f of degree $\leqslant 3$ with $f(0) = 1$ and $f(k) \leqslant 0$ for all integers $k \geqslant 1$; then $\Pr(X = 0) \geqslant Ef(X)$, and $Ef(X)$ is computable from the moments.

Hint 2. Take $f(x) = -\frac{(x-1)(x-2)(x-3)}{6}$. Then $Ef(X) = -\frac{EX^3 - 6EX^2 + 11EX - 6}{6} = \frac{1}{3}$. Equality needs X supported on $\{0,1,2,3\}$: solve for the distribution ($\Pr(X=0) = \frac{1}{3}$, $\Pr(X=1) = \frac{1}{2}$, $\Pr(X=3) = \frac{1}{6}$). The minimum is $\frac{1}{3}$.

Problem 5

Hint 1. Words without 11: states "last letter 0" and "last letter 1", matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$; the count is F_{n+2} and the growth rate $\frac{1+\sqrt{5}}{2}$. For $2 \times n$ domino tilings, cut at the first column: either a vertical domino or two horizontal ones, giving $t_n = t_{n-1} + t_{n-2}$.

Hint 2. For $3 \times n$, use as states the set of cells of the next column already covered by horizontal dominoes sticking out of the current one (8 subsets, but only a few are reachable). Equivalently, t_m for a $3 \times 2m$ board satisfies $t_m = 3t_{m-1} + 2(t_{m-2} + \cdots + t_0)$, i.e. $t_m = 4t_{m-1} - t_{m-2}$, and odd widths give 0.

Hint 3. So the generating function is $\frac{1-x^2}{1-4x^2+x^4}$ in the width n; characteristic polynomial t^2-4t+1 in m; growth rate $2+\sqrt{3}$ per two columns, i.e. $\sqrt{2+\sqrt{3}} \approx 1.932$ per column.